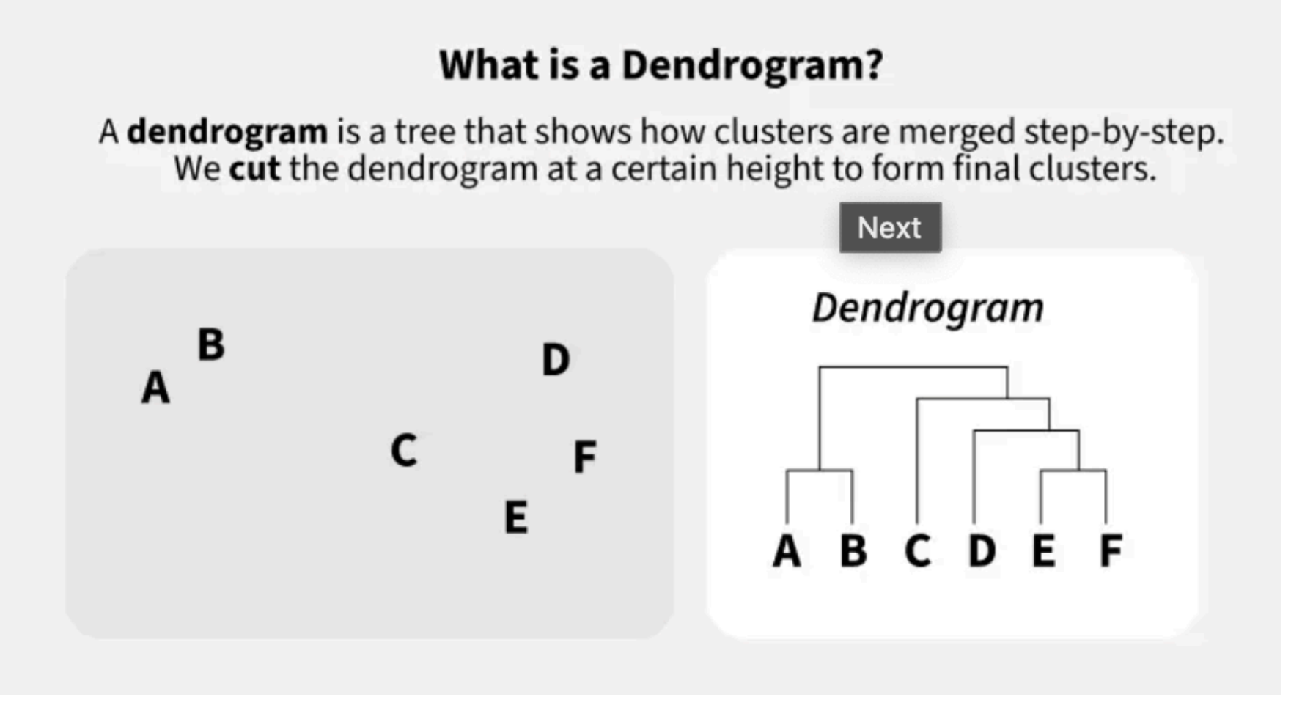
Tutorial 5

Tuesday, 14 October 2025 00:21

HIERARCHICAL CLUSTERING

1. Unsupervised learning technique
2. Grouping of similar data points into hierarchy clusters (tree of nested clusters)
3. Don't need number of clusters specification as clusters are built step by step
4. Cluster analysis finds patters and connections in datasets
5. Shown as a dendrogram diagram



What is a Dendrogram?

A **dendrogram** is a tree that shows how clusters are merged step-by-step. We **cut** the dendrogram at a certain height to form final clusters.

<u>Types of hierarchical clustering:</u>

1. **Agglomerative (Bottom-up)**

- **Each data point considered one single cluster and then they are agglomerated together until all clusters are merged together into ONE (single) clus**ter

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Algorithm: Agglomerative Clustering

Input:
   - Dataset X with n samples
   - Desired number of clusters k (optional)
Output:
   - Cluster labels for each data point
   - Dendrogram showing hierarchical merges

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1. Initialize:
     For each data point xi in X:
        Create a separate cluster Ci = {xi}

2. Compute initial pairwise distances:
     For all cluster pairs (Ci, Cj):
        Compute distance d(Ci, Cj) using chosen linkage method
        (e.g., single, complete, average)

3. Repeat until number of clusters = k (or 1 if building full tree):
     a. Find two clusters (Cp, Cq) with the smallest distance
     b. Merge them into a new cluster Cnew = Cp ∪ Cq
     c. Remove Cp and Cq from the cluster list
     d. Compute distances between Cnew and all other clusters
        Update distance matrix accordingly
     e. Record merge information (Cp, Cq, merge_distance, new_cluster_size)
        → This data is used later for dendrogram plotting

4. Assign cluster labels:
     If a specific number of clusters k is given:
        Cut the dendrogram at height corresponding to k clusters
        Assign points to clusters accordingly

5. Plot results:
     a. Plot data points colored by their assigned cluster label
     b. Use dendrogram to visualize hierarchical merges


End Algorithm
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2. **Divisive (Top-down)**

- **All data is in one single cluster which is split into smaller clusters until each data point is a single individual cluster**

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Algorithm: Divisive Hierarchical Clustering

Input:
   data → set of all data points
   max_clusters → desired number of final clusters

Output:
   clusters → list of final clusters
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1. Initialize:
     clusters ← [data]          # start with all points in one cluster

2. Repeat until number of clusters = max_clusters:
     a. Select the largest cluster (cluster_to_split)
          cluster_to_split ← cluster in clusters with maximum number of points
     b. Remove cluster_to_split from clusters

     c. Apply KMeans clustering with k = 2 on cluster_to_split
          → Obtain two new subclusters: cluster1 and cluster2

     d. Add both new clusters back to the cluster list
          clusters ← clusters ∪ {cluster1, cluster2}

3. End Repeat

4. Return clusters as the final set of clusters


End Algorithm
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AGGLOMERATIVE CLUSTERING

<u>WORKED EXAMPLE 1:</u>

Points	X1	X2
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

Distance Matrix

	P1	P2	P3	P4	P5	P6
P1	0	0.23	0.22	0.37	0.34	0.23
P2	0.23	0	0.15	0.2	0.14	0.25
P3	0.22	0.15	0	0.15	0.28	0.11
P4	0.37	0.2	0.15	0	0.29	0.22
P5	0.34	0.14	0.28	0.29	0	0.39
P6	0.23	0.25	0.11	0.22	0.39	0

- 6 points = 6 starting clusters
- Termination = single cluster

Repeat till termination criteria met:
- *Merge cluster closest*
- *Update matrix*

Distance p1 to p1:
Same point

$$\sqrt{(0.40-0.40)^2+(0.53-0.53)^2}=0$$

P1 to p2 = p2 to p1

$$\sqrt{(0.40-0.22)^2+(0.53-0.38)^2}=0.23$$

P1 to p3 = p3 to p1

$$\sqrt{(0.4-0.35)^2+(0.53-0.32)^2}=0.22$$

P1 to p4 = p4 to p1

$$\sqrt{(0.4-0.26)^2+(0.53-0.19)^2}=0.37$$

P1 to p5 = p5 to p1

$$\sqrt{(0.4-0.08)^2+(0.53-0.41)^2}=0.34$$

P1 to p6 = p6 to p1

$$\sqrt{(0.4-0.45)^2+(0.53-0.30)^2}=0.23$$

New distance matrix:

	P1	P2	P3,P6	P4	P5	
P1	0	0.23	0.22	0.37	0.34	
P2	0.23	0	0.15	0.2	0.14	
P3,P6	0.22	0.15	0	0.15	0.29	
P4	0.37	0.2	0.15	0	0.29	
P5	0.34	0.14	0.28	0.29	0	

*Taking single link distance = minimum distance b/w 2 points

5 clusters
Merge till only 1 cluster left

New distance matrix:

	P1	P2,P5	P3,P6	P4
P1	0	0.23	0.22	0.37
P2,P5	0.23	0	0.15	0.2
P3,P6	0.22	0.15	0	0.15
P4	0.37	0.2	0.15	0

4 clusters

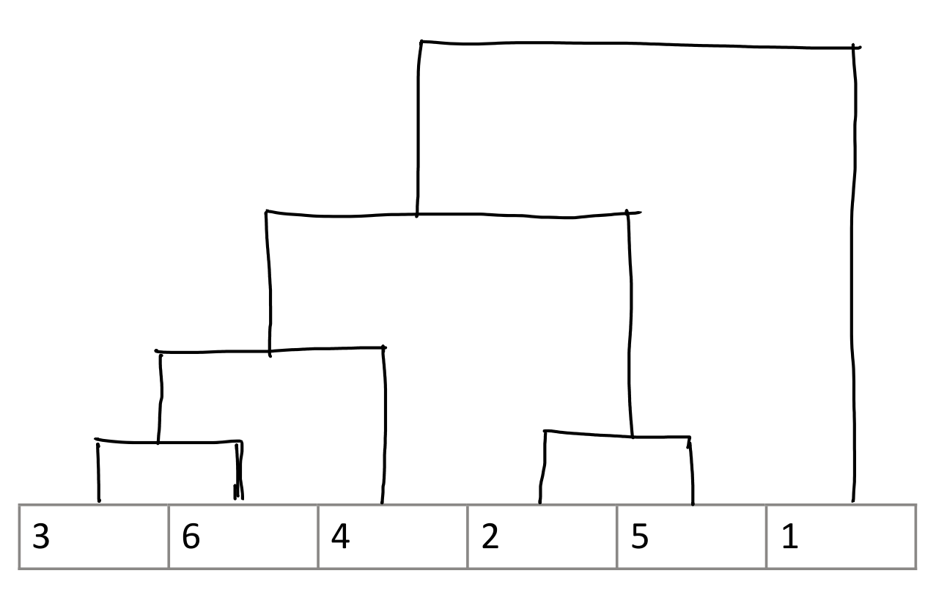
	P1	P2,P5	P3,P4,P6
P1	0	0.23	0.22
P2,P5	0.23	0	0.15
P3,P4,P6	0.22	0.15	0

3 clusters

	P1	P2,P3,P4,P5,P6
P1	0	0.22
P2,P3,P4,P5,P6	0.22	0

2 Clusters

<u>AFTER FINAL MERGER:</u>



DIVISIVE CLUSTERING

<u>WORKED EXAMPLE 2:</u>

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Initial cluster = C1
Contains *a, b, c, d, e*

Separation based on average dissimilarities

1. <u>ab, ac, ad, ae</u>

   1/4*(9+3+6+11) = 7.25

2. <u>ba, bc, bd, be</u>

   1/4*(9+7+5+10)= 7.75

3. <u>ca, cb, cd, ce</u>

   1/4*(3+7+9+2) = 5.25

4. <u>da, db, dc, de</u>

   1/4*(6+5+9+8) = 7

5. <u>ea, eb, ec, ed</u>

   1/4*(11+10+2+8) = 7.75

Highest dissimilarity score is of b and e
Since there is tie doesn't matter what we pick.

C1 = {a, b, c, d, e}
Removing e
New cluster = {e}
Updated C1 = {a, b, c, d}

2nd separation based on average dissimilarities

Updated matrix:

	a	b	c	d
a	0	9	3	6
b	9	0	7	5
c	3	7	0	9
d	6	5	9	0

1. <u>ab, ac, ad</u>

   (1/3*(9+3+6)) - (1/1*(11)) = -5

2. <u>ba, bc, bd</u>

   (1/3*(9+7+5)) - (1/1*(10))=-3

3. <u>ca, cb, cd</u>

   (1/3*(3+7+9)) - (1/1*(2)) = 4.3333

4. <u>da, db, dc</u>

   (1/3*(6+5+9)) - (1/1*(8)) = -1.3333

Highest dissimilarity score is of C

C1 = {a, b, c, d}
Removing c
New cluster = {c, e}
Updated C1 = {a, b, d}

CLUSTER DIAMETER COMPARISON
New cluster = {c, e}
= max{2} = 2

Updated C1 = {a, b, d}
= max{9, 6, 5} = 9

Since 9>2

Separate Updated C1 cluster further

3rd separation based on average dissimilarities

1. <u>ab, ad</u>

   (1/2*(9+6)) - (1/2*(11+3)) = 0.5

2. <u>ba, bd</u>

   (1/2*(9+5)) - (1/2*(10+7))= -1.5

3. <u>da, dc</u>

   (1/2*(6+9)) - (1/2*(8+5)) = 1

Highest dissimilarity score is of D

C1 = {a, b, d}
Removing d
New cluster = {c, d, e}
Updated C1 = {a, b}

CLUSTER DIAMETER COMPARISON
New cluster = {c, d, e}
= max{2, 9} = 9

Updated C1 = {a, b}
= max{9} = 9

Since 9=9

We can separate any of the 2

*If we separate C1 that leaves **a** and **b** alone*

Now separating cluster {c, d, e}

4th separation

1. <u>cd, ce</u>
   1/2*(9+2)=5.5
2. <u>dc, de</u>
   1/2*(9+8)=8.5
3. <u>ec, ed</u>
   1/2*(2+8)=5

Highest dissimilarity D

New cluster = {d}
Updated C1 = {c, e}

5th separation

1. ce
   1/1*(2)=2
2. ec
   1/1*(2) = 2

Any can be seprated